



A LEVEL PHYSICS P1 N2025 MARKING GUIDE

Prepared By Engr T A Mahosi

+263 773 906 073/387

1. A	
2. A	
3. D	
4. C	
5. B	
6. D	➤ halt means the car comes to rest, $v = 0\text{ms}^{-1}$ ➤ from $v^2 = u^2 - 2as$ ➤ $s = \frac{u^2}{2a}$, but $a = \frac{\mu R}{m}$ since Friction is equal to $\mu R = ma$
7. C	➤ Change in momentum, $\Delta p = Ft = \text{Area under } F/t \text{ graph}$ ➤ $mv = \frac{1}{2}bh$ ➤ $v = \frac{bh}{2m} = \frac{(6)(10)}{2(5.0)} = 6.0\text{ms}^{-1}$
8. C	➤ $T = m_2g \dots \dots \dots (1)$ ➤ $T = \text{centripetal force}, F_c \text{ ie } T = \frac{m_1v^2}{r} \dots \dots \dots (2)$ ➤ Equating eqtn 1 and eqtn 2 ➤ $\frac{m_1v^2}{r} = m_2g$ ➤ $v = \sqrt{\frac{m_2gr}{m_1}}$
9. A	➤ $a = -\omega^2x$ ➤ a and x are out of phase hence the phase difference is π
10.D	➤ Light slows down ➤ Frequency is determined by the light source, not by the medium
11.C	➤ $x = \frac{vD}{a}$ therefore $x \downarrow, \gamma \downarrow$ or $x \downarrow; D \downarrow$ or $x \downarrow; a \uparrow$
12.C	➤ Compare the given displacement equation with the general equation of a progressive wave ➤ Given $y = 3.5 \sin(4t + 0.2)$ then from $y = a \sin(\omega t + \phi)$ ➤ It implies $\omega = 4$, ➤ $\frac{2\pi}{T} = 4$ ➤ $T = \frac{2\pi}{4}$

13.A	➤ GaXrUVIMiRa order of increasing wavelength
14.B	
15.A	➤ $n \leq \frac{1}{N\gamma}$ ➤ $\leq \frac{1}{(600 \times 10^{-9})(450 \times 10^3)}$ ➤ $\leq 3.7 \dots$ ➤ $n \approx 3$
16.C	➤ For constructive interference, Path difference = $n\gamma$ ➤ Therefore $n\gamma = 16 - 12$ ➤ $4n = 4$ ➤ $n = 1$ hence $\phi = \pi \text{ rads}$
17.B	➤ When a capacitor is connected to a DC supply, its frequency being zero, and hence the reactance becomes infinite so it will block DC.
18.A	
19.A	
20.B	➤ From ➤ $P = (I_{rms})^2 R$ ➤ $I_{rms} = \sqrt{\frac{P}{R}}$ ➤ $= \sqrt{2}$ ➤ $I_o = \sqrt{2} I_{rms}$
21.B	➤ Radially inwards
22.B	➤ Use Boolean expressions ➤ NAND performs the logical NOT of an AND gate($\overline{A.B}$), and the NOT gate inverts this output effectively canceling out the negation to produce the original AND function ($A.B$)
23.C	➤ From $I = I_o e^{-\frac{\tau}{CR}}$ but $I_o = \frac{V_s}{R_{eff}}$ ➤ $e^{-\frac{\tau}{CR}} = 0.416$ use $R_{eff} = 5000 + 200$ ➤ $V_c = V_s(1 - e^{-\frac{\tau}{CR}})$ ➤ $= 10(1 - 0.416)$ ➤ $= 5.84V$
24.D	➤ LDR has low resistance in light and high resistance in the dark.
25.C	➤ From $E = \frac{Fl}{Ae}$ ➤ If A is halved ➤ $E = \frac{Fl}{\frac{1}{2}Ae}$ ➤ $= \frac{2Fl}{Ae}$
26.B	
27.A	

28.D	➤ Identify the sign conventions according to Thermodynamics, i.e. W on the system is negative and Q given out by the system is negative ➤ $W = -20kJ$ ➤ $Q = -30kJ$ ➤ From 1st Law of Thermodynamics ➤ $Q = W + \Delta U$ ➤ $-30kJ = -20kJ + \Delta U$ ➤ $\Delta U = -10kJ$
29.D	➤ $E_k = mc\Delta T$ ➤ $\frac{1}{2}mv^2 = mc\Delta T$ ➤ $v^2 = 2c\Delta T$ ➤ $v = \sqrt{2c\Delta T}$
30.A	
31.B	➤ $c^2 \propto T$ from translational kinetic energy ➤ $c = k\sqrt{T}$ ➤ If the temperature is halved ➤ $c_1 = k\sqrt{\frac{1}{2}T}$ ➤ $c_1 = \frac{k\sqrt{T}}{\sqrt{2}}$ ➤ $c_1 = \frac{c}{\sqrt{2}}$
32.D	➤ Bernoulli's Principle
33.B	
34.D	
35.B	
36.C	➤ $E = E_2 - E_1$ ➤ $= (-0.85 - -3.40)eV$ ➤ $= 2.55eV$
37.A	➤ Check Binding Energy per Nucleon curve
38.C	
39.A	➤ Einstein's Theory of Relativity ➤ $E = (\Delta m)c^2$
40.C	

By Engr T A Mahosi
+263773906073/+263773906387



Any man who can drive safely while kissing a pretty girl is simply not giving the kiss the attention it deserves.